

Evaluation of the Potential Efficacy of Bacteriophage Therapy for the Management of Pressure Ulcers and Its Comparison to Conventional Therapy: A Randomized Controlled Trial

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 Visual Abstract

 Supplement Material

ABSTRACT

Background and aims: Pressure ulcers (PUs) are common in physical trauma patients who require prolonged stays in the intensive care unit (ICU). Despite standard preventive and therapeutic measures, healing is often a slow process. Conventional systemic antibiotics offer limited therapeutic benefits due to inadequate tissue penetration and the associated risk of driving antimicrobial resistance. Bacteriophage therapy, with its ability to target resistant and biofilm-associated bacteria, offers a potential “alternative.” We aimed to evaluate the efficacy and safety of topical bacteriophage therapy for the management of PUs in critically ill ICU patients and its comparison to conventional therapy.

Patients and methods: This prospective, randomized, double-blind controlled trial was conducted in the ICU of a tertiary care center between April 2024 and May 2025. Seventy-six adult patients with grade II–IV PUs were screened, and 68 were enrolled (34 in each group). Group C received conventional therapy, while group B received conventional plus customized topical bacteriophage therapy. Healing was assessed using the Pressure Ulcer Scale for Healing (PUSH) tool; tissue type, exudate amount, and ulcer dimensions were assessed at baseline and on days 5, 7, 14, 21, and 28. The primary objective was to evaluate the efficacy of bacteriophage therapy for the management of PUs. The secondary objective was to observe the safety profile of bacteriophage therapy.

Results: Both groups demonstrated healing over time, but group B demonstrated significantly greater improvement as compared to group C. The reduction in the mean PUSH score was higher in group B ($p < 0.05$). Group B also showed faster epithelialization, an earlier decrease in exudate, and more closed ulcers. These differences became statistically significant from day 7 onward. No adverse events were reported.

Conclusion: Topical bacteriophage therapy is a safe and effective adjunct to conventional care in the management of PUs among ICU patients.

Keywords: Antibiotics, Bacteriophages, Intensive care unit, Pressure ulcers, Trauma.

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HIGHLIGHTS

- Bacteriophage therapy significantly improved the healing of pressure ulcers (PUs).
- It reduced exudate and promoted epithelialization faster than the control.
- Customized phages targeting patient-specific pathogens, offer a promising adjunct for resistant, biofilm-associated infections in critically ill trauma patients.

INTRODUCTION

Trauma is a modern-day epidemic, with a marked increase in the number of accidents due to the rapid growth of the motor industry and economic expansion. Patients with head injuries, cervical spine trauma, and polytrauma require prolonged admission to the intensive care unit (ICU). Patients in an ICU are at a high risk of developing PUs because of severe trauma and long periods of immobility. Pressure ulcers are considered a major and largely preventable indicator of the quality of medical care. According to the Institute for Health Metrics and Evaluation (IHME), PUs caused 37,000 deaths and 2.4 million new cases worldwide in 2021.¹ A PU is defined as an injury to the skin and underlying tissue over a bony prominence due to pressure or friction.² Pressure ulcers commonly develop on the sacrum, coccyx, heels, and ears. These ulcers can lead to major problems such as infections, longer hospital stays, delayed weaning from ventilatory support, and higher healthcare

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expenses, which affect the patient’s quality of life. In severe cases where there is profound tissue damage, these can be life-threatening.^{3,4} A repositioning strategy is required for every patient who has been diagnosed with a PU or who is at risk of developing PUs. However, once a PU is diagnosed, the management strategy includes pressure relief, debridement, dressings, topical agents, infection control, and reconstructive surgeries. One of the most challenging parts of management is infection control. Infection is one of the leading causes of death after trauma.⁵ Initially, PUs are treated with debridement and topical antiseptics. However,

once infection sets in, these measures become less effective. Even topical antibiotics show reduced action. Antimicrobial resistance is a global public health problem and continues to rise. Bacteria in biofilms are protected by a surrounding matrix, which makes them highly tolerant to antibiotics.⁶ Intravenous antibiotics cannot completely eliminate bacteria inside a biofilm. Poor blood flow and biofilm formation reduces the effectiveness of both systemic and topical antibiotics. Bacteriophage therapy has been used for different types of infected wounds, but no previous studies have specifically evaluated its use in PUs. The primary objective of this study was to assess the efficacy of topical bacteriophage therapy for PUs and compare it with conventional treatment.

PATIENTS AND METHODS

This was a prospective, double blind, randomized study that was conducted in the ICU of an apex care center in the Department of Anaesthesiology and Microbiology. The study period was from April 2024 to May 2025. The Institutional Ethics Committee approved the study protocol. The clinical trial was registered with registration number CTRI/2024/03/064013. Patients admitted to the ICU were screened for eligibility. Inclusion criteria for patient selection were adults (age > 18 years) admitted to the ICU following trauma, including head injury, spinal cord injury, or polytrauma, who developed PUs (grade III–IV). Patients who refused consent, pregnant or lactating women, and patients with systemic diseases (hypertension, diabetes mellitus), dermatological diseases, tumors, burns, or tuberculosis were excluded. The study procedure was explained to all eligible participants and their attendants. Written informed consent was obtained from the patient or from the attendant if the patient was unable to consent. A total of 76 patients were screened for the study. Randomization was performed using computer-generated random numbers. Patients were allocated to either conventional therapy or conventional therapy plus topical bacteriophage therapy. Group allocation was done using sealed opaque envelopes. Group C ($N = 34$) received conventional therapy, and group B ($N = 34$) received conventional therapy with topical bacteriophage application. A baseline assessment was performed, including demographic details, type of trauma, Glasgow coma score (GCS), C-reactive protein (CRP), serum procalcitonin (PCT), total leukocyte count (TLC), acute physiology and chronic health evaluation (APACHE) II score, culture and sensitivity reports from endotracheal or tracheostomy aspirates, blood, and urine; and vasopressor use were recorded. Pressure ulcers were graded using the National Pressure Ulcer Advisory Panel (NPUAP) staging system. Healing was assessed using the Pressure Ulcer Scale for Healing (PUSH) tool. Grading was done before starting treatment. In group C, the ulcer was cleaned with saline. Slough was removed using hydrogen peroxide in large ulcers, followed by a betadine dressing. Debridement and antiseptic application was done once daily. Broad-spectrum intravenous antibiotics were started initially and later modified according to the culture sensitivity report. Treatment continued for 28 days. In group B, the same care was provided along with customized topical bacteriophage therapy. The ulcer was washed with normal saline. Pus and tissue samples were collected using a sterile swab and sent to microbiology for processing and preparation of a customized bacteriophage. Until the customized bacteriophage was ready (usually 4–5 days), a mixture of bacteriophages against isolates most commonly found in the ICU, such as *Escherichia coli*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Klebsiella pneumoniae*, and *Enterococcus faecalis*,

was applied on the PU topically after cleaning with normal saline. The concentration of bacteriophages was 10^9 colonies/mL. The dressing of the PU was done using sterile gauze pieces. Processing of the sample by microbiology included identification of bacterial growth, antibiotic testing, and preparation of customized bacteriophages; this process usually took 4–5 days. After receiving the customized bacteriophage therapy, it was started topically and continued until the 28th day. Wound healing was assessed in both groups on day 5, 7, 14, 21, and 28. The safety of bacteriophage therapy was assessed through continuous clinical monitoring and serial measurement of inflammatory biomarkers, including TLC, CRP, and PCT. Throughout the intervention period, the patient was observed for local adverse reactions (such as erythema, warmth, increased discharge, hypersensitivity, or ulcer deterioration) and systemic adverse events in the bacteriophage group.

Sample Size

Based on the previous study, “Biological therapy on infected traumatic wounds: A case-control study,” the sample size formula for two independent proportions is as follows:⁷

$$n = \frac{(Z_{1-\alpha} + Z_{1-\beta})^2 (p_1 q_1 + p_2 q_2)}{(p_1 - p_2)^2}$$

where $p_1 = 0.74$ (representing a 2-point improvement in depth of wound in 74% of the interventional group)

$p_2 = 0.33$ (representing a 2-point improvement in depth of wound in 33% of the control group)

$Z_{1-\alpha} = 1.96$ at a 5% level of significance

$Z_{1-\beta} = 1.282$ at 90% power

$q_1 = 1 - p_1$

$q_2 = 1 - p_2$

then $n = 29$; further assuming a 15% mortality rate during this period, the final calculated sample size for the study was $n = 34$ in each group.

Statistical Analysis

The data were collected and entered into a Microsoft Excel spreadsheet, MSO 2021. Data analysis was performed using the Statistical Package for the Social Sciences (SPSS) software [International Business Machine (IBM) Corp., Chicago, IL, USA, Version 25.0]. Categorical variables were expressed as numbers and percentages (%), while quantitative data were presented as mean and standard deviation (SD) or as the median and interquartile range. Normal data were analyzed using the Shapiro–Wilk test, and for others, nonparametric tests were used. The quantitative data that were not normally distributed were analyzed using the Mann–Whitney U -test, and normally distributed quantitative data were analyzed using the independent t -test. The comparison of the qualitative variables was analyzed using a paired t -test. If any cell had an expected value of <5 , then Fisher’s exact test was used.

RESULTS

This study was carried out in 76 patients, with 38 participants in each group each. Four patients were excluded from each group as the patients died before the 28 days, and 34 patients in each group were analyzed (Fig. 1).

Baseline demographic and clinical characteristics were well balanced between the two study groups (Table 1). There were

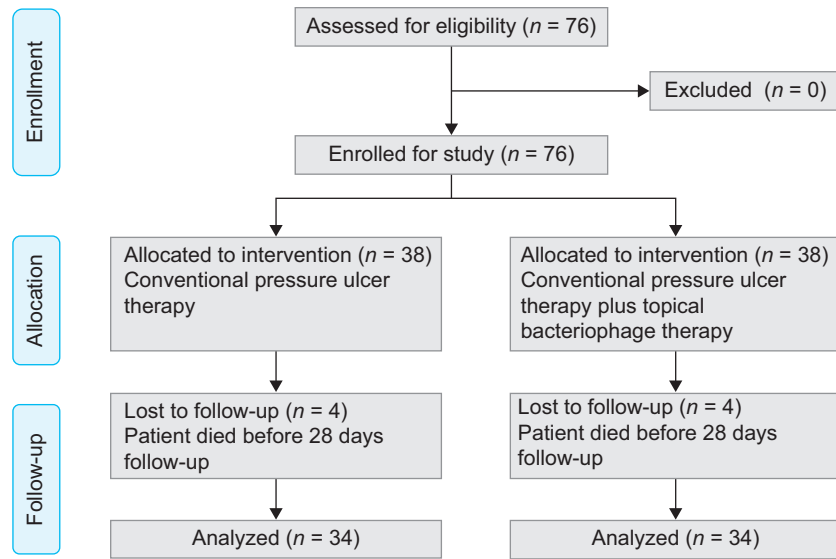


Fig. 1: CONSORT flow of the study

Table 1: Baseline characteristics of groups C and B

Parameters	Group C (N = 34)	Group B (N = 34)	p-value
Age (years)	36.68 ± 7.72	39.56 ± 8.16	0.139
Gender M/F	23 (67.6%)/ 11 (32.4%)	19 (55.9%)/ 15 (44.1%)	0.318
Cervical spine trauma	8 (23.5%)	9 (26.5%)	0.967
Head injury	13 (38.2%)	14 (41.2%)	0.967
Others	8 (23.5%)	9 (26.5%)	0.967
Polytrauma	7 (20.6%)	6 (17.6%)	0.967
GCS			0.009
Mild (13–15)	0 (100%)	0 (100%)	
Moderate (9–12)	4 (11.76%)	6 (23.52%)	
Severe (3–8)	30 (88.23%)	28 (76.47%)	
APACHE II score (% mortality) at admission	15.79 ± 2.47	15.94 ± 3.18	0.832

Data expressed as mean ± SD and as percentages for categorical variables; p-value < 0.05 was considered statistically significant; APACHE, acute physiology and chronic health evaluation; Group B, conventional plus bacteriophage therapy; Group C, conventional therapy; GCS, Glasgow coma score; Gender M/F, male/female; N, number of patients; Others: chest trauma, abdominal trauma, musculoskeletal trauma, maxillofacial trauma

no statistically significant differences with respect to age, sex distribution, type of trauma, or baseline disease severity. Severity of illness at admission, assessed using the APACHE II score, was comparable between groups. Neurological status at presentation, categorized using the Glasgow Coma Scale, also showed no significant intergroup differences, indicating similar baseline neurological impairment. Overall, the comparability of baseline characteristics suggests that the two groups were appropriately matched prior to initiation of the respective treatment strategies.

Table 2 summarizes the baseline distribution of PU grades in both study groups. Only patients with grade III and grade IV PUs were included. The severity distribution of PUs was comparable

Table 2: Baseline grading of PUs in groups C and B

Grading	Group C (N = 34)	Group B (N = 34)	Total	p-value
Grade III PU	27 (79.4%)	20 (58.8%)	47 (69.1%)	
Grade IV PU	7 (20.6%)	14 (41.2%)	21 (30.9%)	
Total	34 (100%)	34 (100%)	68 (100%)	0.066

Data expressed as number of percentages for categorical variables. PU grading was done using the NPUAP pressure injury staging system; <https://npiap.com/page/PressureInjuryStages>; p-value > 0.05 was considered not significant; Group B, conventional plus bacteriophage therapy; Group C, conventional therapy; N, number of patients

between group C and group B, with no statistically significant difference observed between the groups ($p = 0.066$), indicating similar baseline wound severity prior to initiation of treatment.

Figure 2 compares the dimensions of the PUs (length × width) in groups C and B over a period of 28 days, these dimensions were measured at specific intervals (day 0, 5, 7, 14, 21, and 28). Group B (bacteriophage therapy) showed a faster reduction in ulcer size compared to group C. On day 28, most ulcers in group B had reduced to <0.6 cm².

In group C, ulcer size decreased slowly, and many ulcers remained between 2.1 and 4.0 cm² until day 21. The difference between the two groups became clear after day 7. This shows that bacteriophage therapy helped ulcers shrink faster. Group B demonstrated the most effective ulcer healing compared to group C, as evident by the highly significant p-value on day 28 ($p < 0.001$).

Table 3 presents the temporal changes in wound tissue characteristics in both groups. At baseline, no ulcers were closed in either group, and epithelial tissue was minimal. Over the follow-up period, epithelialization progressed in both groups; however, its onset and progression were noticeably earlier and more pronounced in group B. Complete wound closure was observed only in group B, beginning by day 21. Granulation tissue gradually decreased in both groups, with a more rapid decline observed in patients receiving bacteriophage therapy. Similarly, slough reduction occurred earlier in group B and was completely eliminated by day 7, whereas its persistence was noted for a longer duration in group C. From day

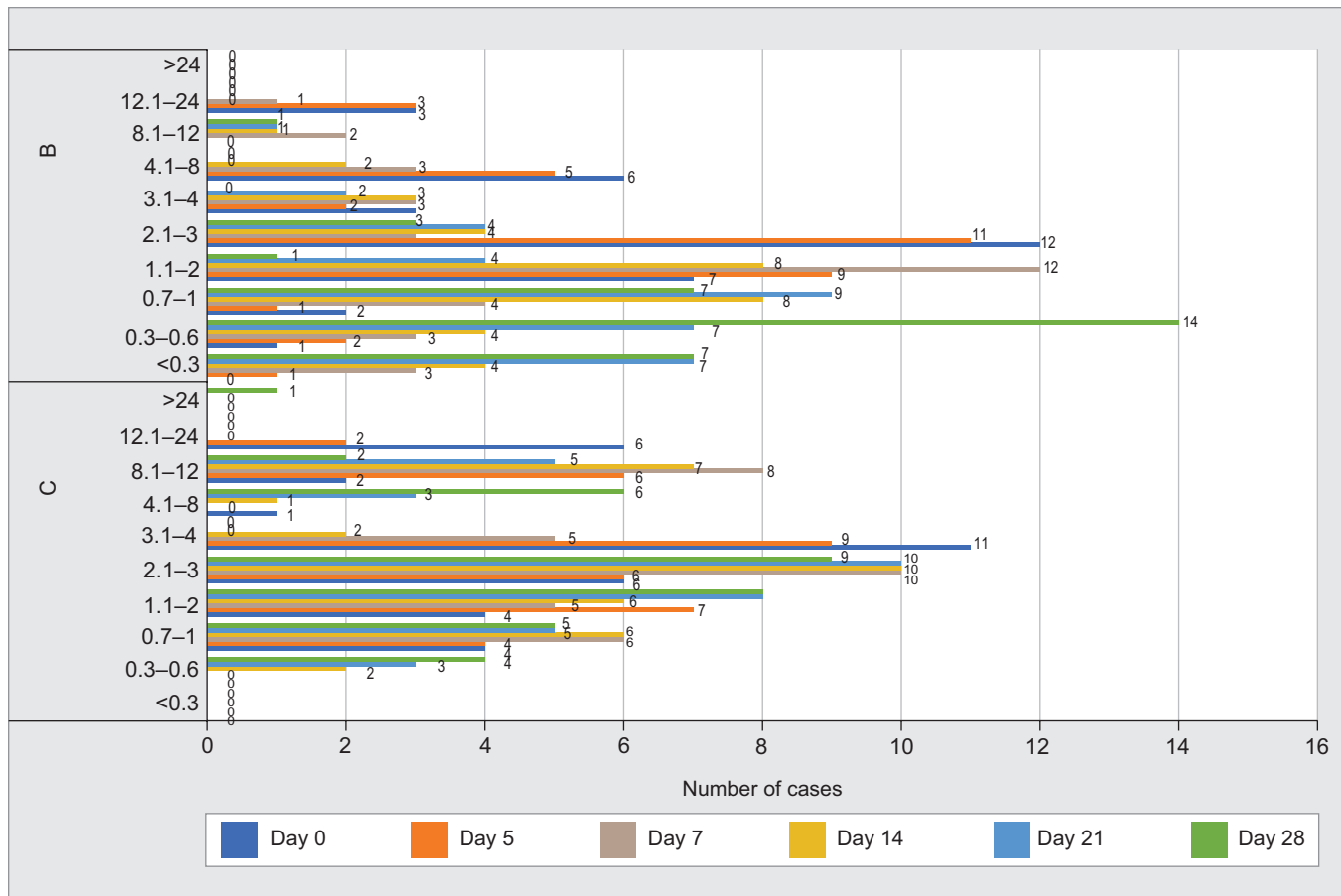


Fig. 2: Comparison of ulcer dimensions (length × width, cm²) between group C (conventional therapy) and group B over 28 days

Table 3: Comparison of tissue type in PUs between group C and B from day 0 to 28

Day	Closed (C)	Closed (B) N = 34	Epithelial (C)	Epithelial (B)	Granulation (C)	Granulation (B)	Slough (C)	Slough (B)	Necrotic (C)	Necrotic (B)	p-value
0	0 (0%)	0 (0%)	0 (0%)	2 (5.9%)	25 (73.5%)	21 (61.8%)	9 (26.5%)	11 (32.4%)	0 (0%)	0 (0%)	0.280
5	0 (0%)	0 (0%)	2 (5.9%)	9 (26.5%)	28 (82.4%)	22 (64.7%)	4 (11.8%)	3 (8.8%)	0 (0%)	0 (0%)	0.070
7	0 (0%)	0 (0%)	6 (17.6%)	13 (38.2%)	24 (70.6%)	21 (61.8%)	4 (11.8%)	0 (0%)	0 (0%)	0 (0%)	0.034
14	0 (0%)	0 (0%)	10 (29.4%)	10 (29.4%)	22 (64.7%)	22 (64.7%)	2 (5.9%)	2 (5.9%)	0 (0%)	0 (0%)	0.009
21	0 (0%)	2 (5.9%)	15 (44.1%)	23 (67.6%)	19 (55.9%)	9 (26.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0.027
28	0 (0%)	2 (5.9%)	15 (44.1%)	28 (82.4%)	19 (55.9%)	4 (11.8%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	<0.001

Data expressed as a percentage for categorical variables; Values are given as a number (percentage); Tissue type was assessed using the PUSH tool; p-value < 0.05 indicates a statistically significant difference between the two groups; Slough and necrotic tissue indicate poor healing; Epithelial tissue and closed ulcers indicate healing improvement; Group B, conventional plus bacteriophage therapy; Group C, conventional therapy; N, number of patients

7 onward, the distribution of tissue types differed significantly between the two groups ($p < 0.05$), indicating faster wound healing dynamics in the bacteriophage-treated group. Overall, these findings demonstrate that adjunctive bacteriophage therapy was associated with accelerated epithelialization and earlier wound closure compared with conventional therapy alone.

Figure 3 compares the amount of exudate between group C and B from day 0 to 28 using the PUSH tool categories (none, light, moderate, heavy). It was measured at specific intervals (day 0, 5, 7, 14, 21, and 28). The X-axis represents the PUSH tool exudate parameters, and the Y-axis represents the number of cases. Color coding indicates the number of days in group C and B; numbers on the bars indicate the number of cases. Group C showed a gradual

reduction in exudate over time, with significant improvements by day 28. On day 28, out of 34 cases, 8 cases had no discharge (none), 15 cases had light discharge, and 11 cases had moderate discharge. In group B, on day 28, 26 cases had no discharge (none), 7 cases had light discharge, and 1 case had moderate discharge. Group B demonstrated a more rapid and significant reduction in exudate on day 28, with most ulcers having no exudate by day 28. The differences between the groups become statistically significant from day 7 onwards, indicating that group B has a more effective reduction in ulcer exudate compared to group C.

Table 4 compares the mean total PUSH scores between groups B and C at baseline and at day 28. At baseline (day 0), the mean PUSH scores were comparable between the two groups, with no

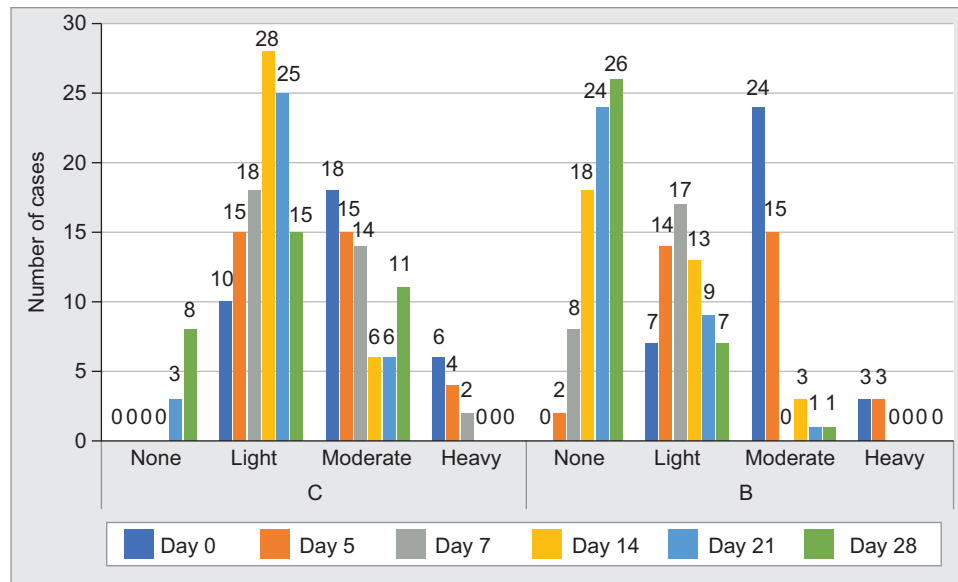


Fig. 3: Comparison of exudate amount between group C and B from day 0 to 28 using PUSH tool categories (none, light, moderate, heavy)

Table 4: Comparison of mean total PUSH score between group C and B on day 0 and 28

Total score	Group C (N = 34)	Group B (N = 34)	Unpaired t-test p-value
Day 0	9.97 ± 2.33	9.44 ± 1.66	0.284
Day 28	7.26 ± 2.38	3.88 ± 2.03	<0.001

Values are presented as mean ± SD; Lower PUSH scores indicate better healing; *p*-value < 0.05 was considered statistically significant; Group B, conventional plus bacteriophage therapy; Group C, conventional therapy; N, number of patients

statistically significant difference observed, indicating similar initial wound severity. By day 28, a significant reduction in the PUSH score was observed in both groups; however, the reduction was markedly greater in group B. The mean PUSH score at day 28 was significantly lower in group B compared with group C (*p* < 0.001), reflecting superior wound healing outcomes in patients receiving adjunctive bacteriophage therapy. These findings indicate that bacteriophage therapy, when added to conventional treatment, was associated with a significantly greater improvement in PU healing over the study period. Comparisons of safety profile in terms of TLC, PCT, and CRP among study groups C and B. A detailed longitudinal comparison of inflammatory and safety biomarkers (TLC, PCT, and CRP) between group B and group C over 28 days is presented in Supplementary Table S1. The progression of wound healing over time in patients receiving bacteriophage therapy (group B) is illustrated in Supplementary Figure S1.

DISCUSSION

In this randomized controlled trial, we evaluated the safety and efficacy of topical bacteriophage therapy in PUs among critically ill trauma patients. Patients in the bacteriophage group (group B) showed faster healing. They had earlier epithelialization, quicker reduction in exudate and slough, and greater improvement in PUSH scores compared to the conventional therapy group (group C). The difference became statistically significant from day 7, matching the expected time for customized phage activity. Previous studies

on phage therapy have focused mainly on chronic wounds or ambulatory patients, such as those with diabetic foot ulcers, venous leg ulcers, and burn wounds. A randomized double-blind trial by Karn et al. reported wound sterilization in 93.3% of cases and complete healing by day 90 with personalized phage therapy, showing its clinical potential.⁸ A systematic review by Narayanan et al. highlighted the potential of phage therapy in wound healing but noted the lack of high-quality controlled trials.⁹ Our study adds evidence in ICU trauma patients, where PUs are deeper, more resistant, and often involve biofilm-forming multidrug-resistant bacteria.

Bacteriophage therapy introduces phages that directly kill pathogenic bacteria.¹⁰ Phages may be lytic, bioengineered, or used as purified lytic proteins. These phages and their enzymes can act against multidrug-resistant bacteria.¹¹ Phages also reduce the risk of antibiotic resistance and cause minimal disturbance to the normal microbiome.¹² Lytic phages selectively target bacteria without harming human cells. Gupta et al. studied bacteriophage therapy in chronic, nonhealing wounds that did not respond to debridement or antibiotics.¹³

They concluded that topical bacteriophage therapy may help treat chronic nonhealing wounds. Bhartiya et al. evaluated bacteriophage therapy in 54 patients with infected traumatic wounds and reported that phage therapy cleared infection faster and was more cost-effective than conventional care.⁷ These studies were conducted in ambulatory patients who already had wounds due to trauma, burns, or infection. In contrast, our study was done in ICU patients. Bhartiya et al. used the Bates-Jensen criteria for wound assessment, whereas we used the PUSH tool. We also monitored PCT and CRP levels, which were not evaluated in earlier studies.

Markoishvili et al. used topical phage therapy in 24 patients with venous stasis ulcers. They applied a commercial phage dressing called PhagoBioDerm and observed improved wound healing in patients who had not responded to antibiotics.¹⁴ They also reported that PhagoBioDerm was safe, well-tolerated, and caused no complications. Safety remains a critical concern when introducing novel biological therapies, particularly in critically ill populations. In our study, safety was evaluated through continuous

clinical monitoring and serial measurement of inflammatory biomarkers, including TLC, CRP, and PCT (Supplementary Table S1). No clinically meaningful elevations in these markers were observed, and no local or systemic adverse reactions occurred in the bacteriophage-treated group. Both groups showed a gradual decline in inflammatory markers over time, reflecting infection control and clinical stabilization. Although a statistically significant reduction in CRP was observed at day 14 in the bacteriophage group, this finding was transient and not associated with adverse effects, further supporting the safety of the intervention. Our findings on safety are consistent with this report. The limitation of their study was the absence of a control group. In contrast, our study was a randomized controlled trial with a control group, and bacteriophage therapy was used in addition to standard treatment.

Jikia et al. also used PhagoBioDerm in patients with radiation-induced, secondarily infected ulcers that were resistant to long-term antibiotic therapy. They reported a reduction in bacterial load, decreased discharge, and visible healing within 2–7 days.¹⁵

Rhoads et al. evaluated the safety of phage therapy using a fixed cocktail of eight phages (WPP-201) against *Pseudomonas* spp., *E. coli*, and *S. aureus* in 39 patients with chronic venous leg ulcers (18 treated, 21 controls).¹⁶ Phage therapy was applied weekly as impregnated dressings for 12 weeks, and patients were followed for wound epithelialization at 24 weeks. Epithelialization occurred in 17 of 21 controls (80.95%) and in 12 of 18 phage-treated patients (66.67%). Microbiological cultures were not evaluated in that study.

In our study, the bacteriophage group showed earlier epithelialization, reduction in slough, and the appearance of closed ulcers by day 21. The difference between groups became statistically significant from day 7. We used customized phage therapy based on culture sensitivity and performed serial cultures from the ulcer. Safety was assessed by monitoring vital signs, local reactions, systemic reactions, and laboratory parameters (TLC, PCT, CRP). No local or systemic adverse events occurred in any patient receiving bacteriophage therapy, consistent with the findings of Rhoads et al.

The efficacy of bacteriophage therapy has also been studied in infected burn ulcers. In a case report by Marza et al., topical phage therapy reduced bacterial colonization on the burn surface, allowing skin grafting within 3 days. Phages were applied using filter paper discs (10^3 PFU) placed directly on the wound.¹⁷ Marza et al. also demonstrated phage multiplication at the wound site without any systemic inflammatory reaction. Our study showed similar results. No adverse effects were observed in our ICU trauma patients, even though they were critically ill and immunocompromised.

In another study involving 30 patients with burn infections caused by multidrug-resistant *P. aeruginosa*, daily bandages soaked in phage solution (10^{10} PFU/mL) were applied for 5–17 days. The authors reported reduced discharge, improved wound healing, and sterile wound bed cultures following therapy.¹⁸ In a randomized controlled trial by Jault et al., 27 patients were divided into two groups, one group received bacteriophage therapy via an impregnated alginate dressing, and the other group received 1% sulfadiazine silver emulsion. It was found that there was a decrease in bacterial load, and this decrease was significant, indicating the efficacy of phage therapy.¹⁹ In one study done by Weber-Dabrowska et al., a combination of topical and oral phage therapy was used in 49 burn patients who failed to respond to antibiotic therapy, and 42 patients (85.71%) had full recovery without any adverse reaction.²⁰ Our findings on safety are similar to the study done by Weber-Dabrowska et al. Bacteriophage therapy has also been used

in diabetic foot ulcers that were infected.^{21–23} In a case series done on six patients, “diabetic foot ulcers” showed improved healing after an average of 7 weeks of topical phage therapy. Topical phage therapy has also been used in uncharacterized chronically infected wounds.

In contrast to earlier studies, our trial was performed in trauma patients, including those with cervical spine injury, head injury, and polytrauma. These patients were critically ill, often requiring tracheostomy and mechanical ventilation. Prolonged ICU stay contributed to repeated infections and the development of PUs.²⁴ No prior study has evaluated bacteriophage therapy specifically for PUs. In comparison with previous studies conducted largely in ambulatory or burn patients, our study uniquely evaluated ICU trauma patients—a higher-risk cohort—and still observed no local or systemic adverse events, with TLC, CRP, and PCT remaining stable, thereby reinforcing that phage therapy is safe even in critically ill populations. Ambulatory patients demonstrate better circulation and lower infection risk, whereas ICU patients are immobile and require frequent repositioning, which increases their susceptibility to PUs. Their longer hospital stay, use of multiple antibiotics, not only adds to the economic burden on the patient but also leads to antibiotic resistance. Beyond just the immediate expenditures of medical care, PUs have a financial impact on quality of life, productivity, and the overall economy. Physicians must therefore discover an alternative method of managing PUs because of the causes elucidated above.²⁵ Improving patient outcomes and lessening the cost burden on healthcare systems can be achieved by addressing this problem through efficient prevention, early detection, and suitable treatment options. The significant expenses linked to PUs can be reduced, as well as the quality of life for vulnerable patients, by healthcare systems placing a higher priority on preventive treatment.

Bacteria that colonize long-standing PUs do so in the form of a biofilm, which leads to failure of traditional antibiotics, adding to the expenditure and increasing both cost and antibiotic resistance.²⁶ The new and improved treatment option for antibiotic resistance and biofilm formation is bacteriophage therapy. Recent research on phage-derived enzymes by Yang et al. has shown their ability to degrade biofilm and extracellular polymeric substances.²⁷ This provides a plausible explanation for our observed faster clearance of slough and exudate in group B, as bacteriophages may have disrupted biofilm architecture, thereby enhancing bacterial clearance. Biofilm resistance mechanisms include the formation of a matrix of exopolysaccharide “increase in expression of stress response gene” expression of stress response gene increases, slowing of cell division, and an increase in anaerobic metabolism with decreased leukocyte activity.²⁸

Studies on bacteriophages have concluded that bacteriophage therapy can counter all these changes and show therapeutic benefits. Innovative phage delivery systems, such as phage-containing hydrogels studied by Abed et al., have also shown promise in accelerating wound healing.²⁹ Our findings align with these mechanistic insights, though we used a direct topical approach, which proved feasible and effective in the ICU setting. It is evident that it is always better to use a combination of antibiotics along with bacteriophage. This was also demonstrated by Khairullin et al., whose study showed that bacteriophages revert resistant bacteria to a sensitive state.³⁰

The primary strength of this study is that it is among the first randomized controlled trials evaluating topical bacteriophage

therapy for PUs in critically ill patients. Its prospective and double-blind design, inclusion of culture-guided customized phage preparations, and systematic assessment of ulcer healing using validated tools (PUSH score, tissue type, and exudate parameters) strengthen the reliability of the findings.

The limitation of this study is that it was conducted at a single center with a relatively small sample size, which may limit generalizability. Follow-up was restricted to 28 days, so long-term outcomes such as recurrence or sustained healing could not be assessed. The process of preparing customized bacteriophages also introduced a time lag before therapy initiation. Future multicenter studies with larger cohorts and longer follow-up are needed to validate these findings.

CONCLUSION

Topical bacteriophage therapy, used in addition to conventional care, significantly accelerated the healing of PUs in ICU trauma patients, as evidenced by improved PUSH scores, faster epithelialization, and an earlier reduction in exudate and slough compared to conventional therapy alone. These findings suggest that bacteriophage therapy is a safe and effective adjunct in the management of nonhealing PUs in intensive care settings.

AUTHORS' CONTRIBUTIONS

Kavita Meena: Conceptualization, Software, Project administration, Writing—Review and Editing. Rajesh K Meena: Investigation. Sarita Kumari: Writing—Original Draft, Writing—Review and Editing. Premkumar T: Data curation, Writing, Original draft preparation, Investigation. Gopal Nath: Supervision.

SUPPLEMENTARY MATERIALS

The supplementary table and figure are available on the website www.ijccm.com.

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